CS 438-538, Spring 2025, Hong

Project 3 Internet Measurements

In this project, you will use a few tools to measure network performance of your own daily on-line environments. You should decide for yourself on what you'd like to measure so to choose tools that serve your goal the best. You should also choose your network connections, and metrics you want to study. You will then report your goal, methods and results.

This description has multiple parts as guidelines. You will need to explore what each tool can measure and how to use them for your own project goal. The knowledge learned from Chapters 1 - 5 is very helpful in completing this project. The requirements for the project deliverables are given below.

Measurement examples discussed in classes are included in the ppt posted with this file.

(1) Tools (note: OS dependent): iperf3, ping, tracert (traceroute), ipconfig (ifconfig), dig, telnet (SSH).

(2) What to measure and which tools to use: You must measure at least three performance metrics using iperf3 and another tool picking from the above. Each of the two tools needs to have unique contributions to the measurements. They are complementary. Note, usually each tool gives you information items that satisfy multiple metrics, or say, observations. So, cross references are possible when using multiple tools. The project also requires the measurements to be performed for three different conditions for the same performance goal, i.e., you repeat the measurements for three different cases. The possible conditions are given in (3).

(3) Set up measurements: your connections could be: wired, wireless, on or off campus, mobile or lab/office bound, with or without VPN, noticeable different destinations (e.g., your own machine, on campus, with long distance, etc.), and potentially the gateway you use to access internet, or, an iperf3 server running for the measurements.

(4) Document results: you will use tables, figures, or drawings to show your measurement scenarios and results. The tables and/or figures should be done in a way that comparisons are made clear.

(5) Write a report of 3 pages and submit to BBL. Report should be in ACM or IEEE conference format (e.g., two columns, single space, 10 or 11 font size, the max size of a figure /drawing be 1/6 of the page size, embedded). The filename should be *FirstnameLastname.pdf*, or, *FirstnameLastname.doc.*

A summary on what you will produce relating to the above guidelines.

(0)	What do you measure; And an overview of what you do and results
(1), (2)	The tools you use and why you choose them
(3)	Include your own machine's IP for all the measurements. Describe your connections and draw a topology. For each case per (3).
(4), (5)	Show the full command of the tool you choose (e.g., the command line and input parameters, such as `ping ua.edu`) for each results obtained; put results in tables or figures if possible; explain what you observed with each commend; cross verify the measurements; make comparison across the three conditions; identify the differences **and explain why**. If a result comes out as a long list, make it as an appendix. You can put multiple results side-by-side in an appendix. Extract major observations into a table or a figure.
(6)	Be correct, clear, concise, and logically flow smoothly. A polished college-level report. Grading is based on the mastering of the tools; the completeness in design, execution, and understanding the results; and the quality of the written report. Formatting: You name, and project title. Within three pages. Font size 11 or 10, single space, figure/table maximum size be 1/6 page, embedded in-text.

A table could look like below. But feel free to have your own format. To show the connections of your test cases, you should using separate drawings. localhost (120.0.0.1)

WiFi only	Net set 1 (condi 1)	Tool 1 iperf3	Tool 2 ping	Comparisons
	Metric 1 packet loss	0%	0%	
	Metric 2 jitter (ms)	0.041 ms	1.118034	
	Metric 3 no. of datagrams	21	4	
WiFi with Cisco VPN	Net set 1 (condi 2)	Tool 1	Tool 2	
	Metric 1			
	Metric 2			
	Metric 3			
	Net set 1 (condi 3)	Tool 1	Tool 2	
	Metric 1			
	Metric 2			
	Metric 3			